Minerva (2025) 63:535–555
https://doi.org/10.1007/s11024-024-09547-x

Participatory Convergence: Integrating Convergence and Participatory Action Research

Laura Castro-Diaz[1] · Anais Roque[2] · Amber Wutich[3] · Laura Landes[4] · WenWen Li[5] · Rhett Larson[6] · Paul Westerhoff[7] · Mariana Marcos-Hernandez[8] · Mohammad Jobayer Hossain[3] · Yushiou Tsai[9] · Ramon Lucero[10] · Griffin Todd[11] · Dave White[12] · Michael Hanemann[13]

Accepted: 17 September 2024 / Published online: 18 November 2024
© The Author(s) 2024

Abstract This paper introduces the concept of "Participatory Convergence" as a framework to meet grand social-ecological challenges. Participatory Convergence combines the principles of Convergence Research with Participatory Action Research (PAR), offering a novel approach to tackling complex societal problems. Convergence Research seeks to foster high-level integration between diverse disciplines to address multifaceted issues, emphasizing systems thinking and solutions orientation; however, existing literature falls short in providing practical models for the deep integration of diverse disciplines, community partners, and community members. This paper aims to bridge these gaps by integrating Convergence Research with PAR. We illustrate the application of Participatory Convergence with a case study: the "Action for Water Equity" project, focusing on water challenges faced by communities in U.S. colonias along the U.S.–Mexico border. The Action for Water Equity project is

This paper was written in collaboration with the Action for Water and Equity Consortium. The list of the consortium co-authors can be found at the end of the manuscript.

✉ Laura Castro-Diaz
Laura.CastroDiaz@umb.edu

Amber Wutich
Amber.Wutich@asu.edu

[1] School for the Environment, University of Massachusetts, Boston, MA, USA

[2] Nicholas School of the Environment, Duke University, Durham, NC, USA

[3] School of Human Evolution and Social Change, Arizona State University, Tempe, AZ, USA

[4] Rural Community Assistance Partnership, Washington, DC, USA

[5] School of Geographical Sciences and Urban Planning, Arizona State University, Tempe, AZ, USA

[6] Sandra Day O'Connor College of Law, Arizona State University, Tempe, AZ, USA

a practical example of how Participatory Convergence can be applied to tackle pressing challenges while embracing diversity, inclusivity, and adaptability. This approach holds the potential to facilitate comprehensive solutions for global challenges and foster meaningful change through interdisciplinary collaboration, community engagement, and a commitment to sustainability and equity.

Keywords Participatory convergence · Participatory action research · Water security · Knowledge co-production · Colonias

1 Introduction

Grand challenges such as flooding, droughts, heat, resettlement, mortality, and food security, among others, under climate change pose an existential threat to humanity (IPCC 2022). Confronting these interconnected challenges requires integrative approaches and responses, including cross-disciplinary research integrated with the non-academic sector (Apgar et al. 2009; Iwaniec et al. 2020; Jacobson and Robinson 2009; Lang et al. 2012; Moewaka Barnes et al. 2021; O'Rourke et al. 2016; Sheppard et al. 2011; Sillitoe 2004; Stock and Burton 2011; Wilson 2019). This paper seeks to build on past innovations and research approaches to address this need by introducing "Participatory Convergence" (Moran et al. 2022; Peek et al. 2020; Roque et al. 2022). A problem-based research framework that creates respectful partnerships with stakeholders, integrates disciplines, and enhances knowledge through action and reflection to pursue just and meaningful solutions to societal problems.

Convergence Research is a novel approach that builds on prior innovations like multidisciplinary, interdisciplinarity, and transdisciplinary research (Apgar et al. 2009; Lang et al. 2012; Wickson et al. 2006). In contrast to other approaches, teams engage in Convergence Research to co-create new frameworks, methodologies, or even fields of study to explore complex questions derived from society's vexing problems to create solutions (Bainbridge and Roco 2016; NSF 2019; Peek et al. 2020). Convergence Research aims to achieve high levels of integration between

[7] School of Sustainable Engineering and the Built Environment, Ira A. Fulton Schools of Engineering, Arizona State University, Tempe, AZ, USA

[8] School of Sustainable Engineering and The Built Environment, Arizona State University, Tempe, AZ, USA

[9] ByWater Institute, Tulane University, New Orleans, LA, USA

[10] Rural Community Assistance Corporation (RCAC), West Sacramento, CA, USA

[11] Communities Unlimited, INC, Fayetteville, USA

[12] Global Institute of Sustainability and Innovation, Arizona State University, Tempe, AZ, USA

[13] School of Sustainability, Arizona State University, Tempe, AZ, USA

diverse and distant fields of study (e.g., natural sciences and social sciences) to transcend traditional boundaries and tackle complex problems (National Research Council 2014; O'Rourke et al. 2016; Wilson 2019).

A distinguishing element of Convergence is the focus on systems thinking and solutions orientation (Bainbridge and Roco 2016; Gajary et al. 2023; Peek et al. 2020). Convergence researchers recognize the need for better integration of social sciences and humanities to approach complex challenges rooted in social, political, and ethical conflicts (National Research Council 2014; O'Rourke et al. 2016; Peek et al. 2020; Pohl et al. 2008) and to include and converge with experts from multiple disciplines, private sector, and those affected by the challenges (National Research Council 2014; O'Rourke et al. 2016; Roco 2016). For example, recent efforts to converge local communities, social sciences, engineering, and communication scholars have been made to address renewable energy to support the Sustainable Development Goals (Moran et al. 2022). Yet, models and methods for deep integration of diverse disciplines, community partners, and community members are finite in Convergence literature. This article aims to bridge the gaps in the Convergence literature by integrating it with Participatory Action Research (PAR). We introduce Participatory Convergence as a framework where scholars from different fields, community partners, and members collaborate to solve current societal challenges. We delve into a case study: "Action for Water Equity," focusing on communities located on the U.S-Mexico border in Texas, New Mexico, Arizona, and California called Colonias and their water challenges.

First, we introduce Convergence and Participatory Action Research and then integrate them into Participatory Convergence. The remainder of the paper is focused on describing a case study, the Action for Water Equity project, through the lens of Participatory Convergence. This example is not meant as a blueprint but as an illustration of one form a Participatory Convergence study can take.

2 From Convergence to Participatory Convergence

The early Convergence literature strongly focused on nanotechnology, biotechnology, and new technology based on cognitive science (Roco et al. 2013; Roco and Bainbridge 2003). Later, more focus was given to social and ethical implications given its value for society application (Bainbridge and Roco 2016; Roco and Bainbridge 2003). As an approach to transdisciplinary integration, Convergence brings a unique opportunity for team science to address complex challenges (Bainbridge and Roco 2016; NSF 2019; O'Rourke et al. 2016; Peek et al. 2020).

The goal is to bring together intellectually diverse researchers to effectively communicate across disciplines by adopting common agendas and methodologies as well as propose solutions by developing novel ways of thinking about research questions and outcomes (Bainbridge and Roco 2016). This evolving and non-linear process requires reciprocal exchanges between fields and forms of knowledge to create added value and achieve shared objectives (Bainbridge and Roco 2016). Convergence challenges traditional silos of knowledge production, fostering a more holistic understanding of phenomena.

Convergence Research has recently been adopted for environmental, climate change, and disaster research. For example, scholars from the hazards and disaster field are strongly interested in the Convergence agenda, mainly focusing on collaboration across disciplines (Peek et al. 2020). New Convergence studies aim to manage mega-fires risk (Bukovsky et al. 2022), improve urban resilience to disasters (Chester et al. 2021), enhance water security in high-income economies (Wutich et al. 2022), reverse the salination of freshwater (Grant et al. 2021), and design innovate energy solutions (Brown et al. 2022; Harper et al. 2023; Moran et al. 2022).

There is a need for approaches that investigate human dimensions of grand challenges for Convergence Research. Human-centered design is one approach that has been suggested for incorporating human perspectives into Convergence Research (Gajary et al. 2023). Human-centered design is particularly valuable for Convergence Research that foregrounds technological feasibility, economic viability, and the needs of humans as "users," often in the contexts of product development or technology uptake (Chen et al. 2020). However, some grand challenges require approaches with the capacity to understand and act on social, cultural, and political systems. For such Convergence Research, participatory approaches may be more appropriate (Roque et al. 2022; Gubrium and Harper 2016). Participatory approaches, in contrast to human-centered design, foreground partnerships, empowerment, and the needs of humans in communities, often in academic or social action research (Chen et al. 2020). Here, we discuss Participatory Convergence as a suitable approach for addressing grand challenges have inextricable social, cultural, and political dimensions.

Participatory approaches, such as Participatory Action Research (PAR), bring new perspectives and opportunities to support the convergence of engineering, biophysical science, and social sciences—while emphasizing the co-creation of knowledge with stakeholders, including those affected by the societal challenge (Bradbury et al. 2019; Cornish et al. 2023; Díaz-Arévalo 2022; Vaughn and Jacquez 2020). PAR aims to investigate reality in order to transform it (Fals Borda 2006). Conducting research as a team that includes stakeholders, such as community members, organizations, and researchers, is an essential aspect of PAR (Greenwood and Levin 2007). The level of participant involvement varies across approaches, including community-based research, citizen science, and participatory rural appraisal (for a more comprehensive review, see Cornwall and Jewkes (1995); Vaughn and Jacquez (2020), and Roque et al. (2022)).

PAR emphasizes the importance of conducting research *with* people to drive social transformation (Díaz-Arévalo 2022). It recognizes multiple ways of knowing and values the insights of diverse perspectives (Fals Borda and Rahman 1991), particularly those of marginalized or oppressed groups. An empowering aspect of PAR lies in its ability to foster more democratic forms of knowledge; that is, in PAR, knowledge generation is not just about progress or knowledge for its own sake, but it is a process that generates knowledge for action and, through action, serves the goals of specific communities (Cornish et al. 2023; Gaventa and Cornwall 2015; Reason and Bradbury 2008). This is achieved by mobilizing groups of people to act on their own affairs while also engaging in critical reflection and learning (Gaventa and Cornwall 2015). Therefore, it is considered liberating and emancipatory and

can lead to significant progress in addressing societal issues (Reason and Bradbury 2008).

Together, Convergence Research and PAR offer a framework that acknowledges the complexity and multiplicity of realities. Integrating diverse forms of knowledge and engaging with stakeholders in the research process facilitates a more nuanced understanding of social phenomena. This epistemological stance recognizes that knowledge is not static but is constructed through dialogue, collaboration, and reflection.

Participatory Convergence integrates the key features of Convergence and brings PAR to support it by involving the active participation of stakeholders. Participatory Convergence brings the two critical features of Convergence: research motivated by (1) a specific problem ("problem-based"); and (2) deep integration of different disciplines (NSF 2019). Participatory Convergence then integrates Convergence with PAR's three key features: (1) partnership, (2) reflexivity, and (3) actionability & significance (Bradbury 2010) (Fig. 1).

3 Water Equity Through the Lens of Participatory Convergence

In this section, we present the Action for Water Equity project as an illustration of the Participatory Convergence approach. The project aims to converge around a solution for water security in U.S. colonias by testing how to integrate engineered

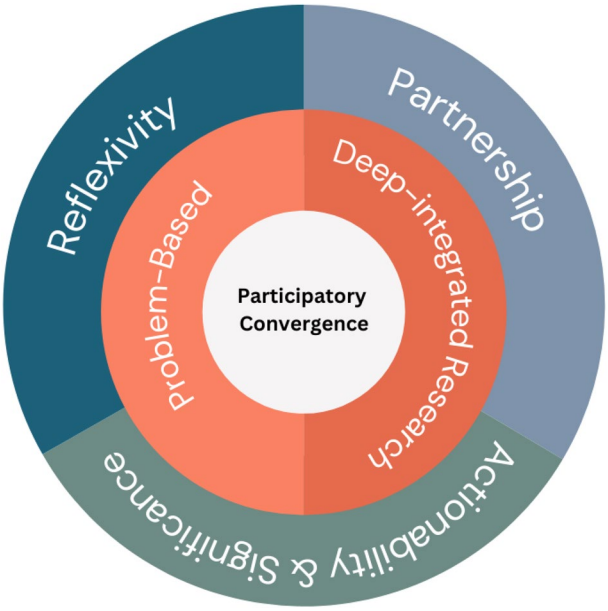


Fig. 1 Participatory Convergence framework. Convergence features (Problem-Based, Deep Integrated Research) may be advanced through a PAR engine (Partnership, reflexivity, and actionability & significance)

and social water infrastructure. We will then leverage our findings to inform solutions for communities facing similar water challenges globally.

3.1 Problem-Based: Water Insecurity

Convergence is problem-based research that aims to address the most urgent societal problems (Moran et al. 2022; Peek et al. 2020; Roco and Bainbridge 2013). Then, Convergence Research is driven by a specific problem (NSF 2019), in our Action for Water Equity project, the problem that drives our research is water insecurity in U.S. colonias. Water insecurity is the inability to access safe and reliable water for human well-being (Jepson et al. 2017; Wutich and Ragsdale 2008). Over half of the world's population experiences water insecurity in the form of severe water scarcity for at least one month annually (IPCC 2022). In the U.S., over one million people lack drinking and sanitation water (Meehan et al. 2020; Roller et al. 2019). Water insecurity disproportionately impacts marginalized populations (J. Brown et al. 2023), like the case of U.S. colonia residents.

Colonias are home to up to one million people, many of whom have limited or no access to drinking water (Jepson 2014; Landes et al. 2021; Wutich et al. 2022). Colonias—which are under-resourced communities—have historically suffered from political exclusion, municipal underbounding, lack of legal resources, and other challenges, resulting in water insecurity (Wutich et al. 2022; Jepson and Vandewalle 2016; Jepson and Brown 2014; Jepson 2012)

Colonia residents face challenges in acquiring affordable, safe water services—all critical to human health, economic well-being, and environmental sustainability (Zheng et al. 2022). When not received from formal sources, colonia residents get water from their own unregulated wells and infrastructure, which poses risks for mosquito infestations, harmful pesticides, and high levels of arsenic (Chahin 2005; Juarez et al. 2022). In the absence of water lines and wells, residents rely on hauling water—a time-consuming, expensive, tiresome task, and poses a range of contamination and mental health risks (Jepson 2014; Landes et al. 2021). Households in these areas often struggle to decontaminate their water collection barrels. Some treatments promise solutions such as small modular and decentralized water treatments (e.g., point-of-use (POU) systems); however, these are not yet well integrated with critical social infrastructure (e.g., law, economy, and cultural water-sharing norms) (Stoler et al. 2022; Vandewalle and Jepson 2015).

In addition to the above, legal uncertainties and potential liabilities complicate the jurisdiction of regulatory structures such as the Safe Drinking Water Act. Ownership of water rights, compliance with applicable permitting, water quality, and certification requirements also pose challenges to implementing solutions (RCAP et al. 2015). The lack of legal representation to resolve these issues further complicates the matter. The creation of social infrastructure to deliver, build, and maintain small-scale water treatment technology is not just an economic or engineering challenge; it requires the comprehensive integration and coevolution of social and physical infrastructure to promote economic prosperity and overcome water insecurity (Wutich et al. 2023).

Thus, the clean water access challenge in the U.S. requires the convergence of diverse scientific methods, traditions, and approaches to facilitate better social and engineered infrastructure integration. As such, the Action for Water Equity team is working with residents from 64 colonias and community partners to bring an opportunity to co-develop solutions that can serve these and other water-insecure communities.

3.2 Deep-Integration of Disciplines

Solving the most urgent societal problems requires integrating multiple disciplines (O'Rourke et al. 2016) (e.g., social science, natural science, engineering, etc.), epistemologies, and backgrounds. One key aspect of Participatory Convergence projects is deep-integrated research; it includes the involvement of scholars from different disciplines and different stages of their careers from diverse backgrounds and affiliations. In our project, we developed deep integration across disciplines and partnerships through consultation, collaboration, and empowerment to ensure that all relevant stakeholders were included in our process. Our team is comprised of researchers from six different fields: anthropology, data science, engineering, economics, law, and policy, along with community partners such as the Rural Community Assistance Partnership (RCAP), Rural Community Assistance Corporation (RCAC), and Communities Unlimited (C.U.), and residents from 64 U.S. colonias.

Participatory Convergence research is unique in its approach to building a team of researchers who share a common culture. Team members start with a solid understanding of their own disciplines but then work to develop the ability to communicate and understand each other's disciplinary language (Gajary et al. 2023). To move from un-converged to converged understandings of our research objectives—including the problems, causes, and solutions—we rigorously monitored our progress using various self-study methods to ensure that every team member interacted and learned from each other. We developed several tools to increase our collaboration and integration; these included monthly speaker series with water experts and designing education tools such as story maps and engineering lessons. We also developed a consortium co-authorship model so academics and community partners could all contribute to and be recognized in publications. Additionally, as one goal of Participatory Convergence is to create a new language that transcends traditional disciplinary boundaries that can be shared across all fields (Gajary et al. 2023; NSF 2019), we have developed specific mechanisms to facilitate this process and promote Participatory Convergence in different stages, such as "Value Proposition" and "Participatory Convergence Data Science."

3.2.1 Value Proposition Framework

Value Proposition allowed us to share and discuss the project from the diverse perspectives of the team members (Westerhoff et al. 2021). We conducted biweekly meetings in which a team member explained their views on the Need, Approach, Benefits, and Competition ("NABC" for short) of our shared effort (See Table 1).

Table 1 Needs, approach, benefits, and competition (NABC) value proposition framework

Step	Brief description
Needs	Identify the vital end-user and societal *needs* that draw your attention and are necessary to make a societal impact. You have to identify the crucial insight of this unmet need with its underlying causes.
Approach	Determine a unique, compelling *approach* for addressing the need. To provide a successful solution, you must first identify the critical insight and offer a sustainable model that provides a competitive advantage. Your solution should fulfill a need, be affordable, easy to use, and meet the demand in ways that end-users will want to adopt.
Benefits	To ensure the best results, assessing the specific quantitative *benefits is important*. This should be done in comparison to all current and foreseeable alternatives, focusing on improving the user experience and reducing costs in multiple dimensions. Identifying and mitigating any significant risks and providing superior benefits to all stakeholders is essential.
Competition	Determine how these benefits quantitatively surpass other alternatives, taking into account various factors such as technology trends, business models, regulations, and demographics. You should be able to recognize your current and future competitors and showcase your strengths. To establish a sustainable competitive advantage, you should analyze both direct substitutions and alternatives for both current and future scenarios.

Then, everyone had a chance to give feedback on the pitch and highlight its strengths and weaknesses. By using this approach, several discussions about the project emerged. For instance, the process of articulating and iterating on the "Need" helped overcome siloed biases and identify transformative approaches. Another significant topic that arose while using the Value Proposition framework was how incorporating more metric-based information could enhance potential solutions for colonias and identify directions for further integration. A data science researcher describes the advantages of using this framework:

> The data collected from our team's NABC presentations indicates Convergence evolution, and we can observe that as time passes, many researchers learned significantly from previous researchers' talks and started to think about how to apply new knowledge and expertise to their own research interests. This sharing of expertise is how we achieved Convergence.

3.2.2 Participatory Convergence Data Science

We developed a deep learning-based semantic analysis to track and measure our Participatory Convergence among team members with diverse expertise over time (Li 2020; Li et al. 2014). We utilized Natural Language Processing (NLP) to analyze group discussion transcripts from Value Proposition presentations. This process involved extracting the main ideas and determining the semantic similarity between team members' ideas. By doing so, we could track the flow of ideas from one team member to another and obtain a real-time understanding of how the team Convergence was achieved over time. For example, we could identify which sub-team

started to converge earlier than the others. A community partner describes how tracking the group interaction allowed a deep Convergence:

> To see if our team's approach is effective, we needed a way to assess the scientists' progress in converging with community partners and members. By combining our Participatory Convergence and Scientists' Self-Study Convergence data, we could track progress over time in shared understandings in how we can address water insecurity.

Figure 2 shows how, over time, the team adopted concepts from other fields and disciplines, creating a common language and converging perspectives and ideas. To illustrate, in the Summer of 2021 one of the team's water law researchers led the first value proposition activity and introduced to the team the concept of building "social infrastructure." He noted:

> …our approach is to try and bring in all of the different areas of expertise that we can, to be able to have something tailored to improve water services to these. This includes improve **social infrastructure** and Community Engagement to make sure that the expertise of the residents are there, that their voices heard that they're giving us the information.

This concept was later adopted and used by other team members, such as anthropologists, Biological scientists, and community partners (See Figure 2). Here we show how the concept "social infrastructure" was used by an anthropologist later in the Fall of 2021:

> Speaking from the social science team, our approach is generally one of developing participatory convergence, and there are three pieces that go into this. The first is understanding high-level political drivers of water insecurity in colonias. These include political exclusion, immigration policy, and economic

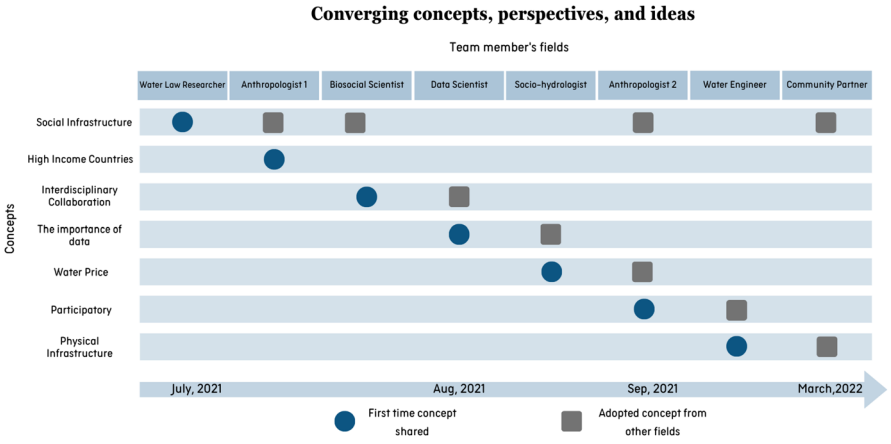


Fig. 2 Flow of team members sharing discipline-specific concepts and perspectives that were eventually adopted by others

> marginality. Within that, we also focus as researchers on community-level forms of **social infrastructure** that can meet water needs, even in the absence of resolving these high-level political problems that are the ultimate cause of water insecurity in colonias. So, the **social infrastructure** we believe that we can help to advance, to meet water needs are informal economic mechanisms supporting cultural norms that produce water sharing and enhancing social networks that can enact solutions to water insecurity.

Other ideas such as "participatory," "interdisciplinarity," and "physical infrastructure" were also interchanged among members of the team, as well as the importance of considering aspects such as "data" and "water price." These concepts were also used by community partners, which reflects the effectiveness of the team collaborative approach.

3.3 Partnership

A key aspect of PAR is the importance of researching *with* and not *on* people. PAR focuses on the quality of relationships with stakeholders and their involvement throughout projects (Bradbury 2010), aiming to address issues significant for co-researchers (Reason and Bradbury 2008). Participation here refers to the level of empowerment local actors have in making decisions (Arnstein 1969); it helps create successful strategies and encourages stakeholders to take ownership of policies and projects (Pretty 1995). The degree of stakeholder engagement, empowerment, and consequent participation can fluctuate among and within individual projects (Cornwall 2008; Vaughn and Jacquez 2020).

Participatory Convergence actively seeks to engage participants as collaborators throughout various stages of the project (Lakhina et al. 2021; Roque et al. 2022). As mentioned above, in our project, besides academics, two main groups of stakeholders are integrated into our group: community partners and local residents.

3.3.1 Community Partners

Our first and most crucial task was establishing a partnership between stakeholders and scholars from different disciplines interested in knowledge integration, co-learning, and co-creation to address water insecurity. As a university team committed to our local community, we built partnerships designed to address water challenges in the U.S. Southwest. Our partners—RCAP, CU, and RCAC—were invited to participate in the research project at the proposal development stage. We interacted and had conversations with colonia residents thanks to the assistance of our partners, who have a long-standing, trustworthy relationship with colonia residents. To illustrate, a community partner noted:

> As part of an organization that works directly with communities, it is refreshing to see a research team commit to and follow through with community-engaged work. At RCAP, we strongly believe if we are not involving commu-

nity members from the very beginning, we will completely miss the point on how to address water insecurity effectively and sustainably.

3.3.2 Local Residents

Developing collaboration skills is critical for PAR projects, including conducting research, sharing results, and facilitating activities (Cornish et al. 2023). The academic team, led by collaborators with expertise in participatory methods, trained and capacitated itself to work in partnership with communities. This included weekly meetings with community partners, self-study sessions in which we interrogated our assumptions about what was "useful" or "valuable" as a solution, and we invited guest speakers to educate and advise us about how to work respectfully together. We centered respect, trust, transparency, engagement, and sensitivity to power imbalances as core concerns (Banks and Brydon-Miller 2018; Boser 2007; Israel et al. 2005; Thomas and O'Kane 1998). While the academic team's work with community partners and colonias communities had many challenges—including over budgets, ways of engaging the colonias during the COVID-19 pandemic, and project goals—these core ethical commitments led us to listen closely and be flexible in working together, as described by a colonia resident: *"I certainly hope that outside experts will help with our community's water challenges. Proper engineering and outside financial resources could help us a lot."*

Due to the outbreak of COVID-19, our team couldn't visit the communities in person as planned. And many communities lacked access to the Internet, making it challenging to conduct team-wide meetings. We developed an approach to incorporate the viewpoints and expertise of local residents into our project: Community-Based Participant Observation (CBPO) (Roque et al. 2023). CBPO is an approach, inspired by the work of Fals Borda (Fals Borda 1979) and others, in which residents are trained to conduct participant observation (Bernard 2017) in their own communities and then document and share the data with academic partners. Table 2 displays a step-by-step guide on how to create a protocol for conducting CBPO.

Our interactions with our partners included engagements with many sub-teams: academic partners working weekly with our community partners; community partners working with colonias residents; and meetings where all worked together to discuss and learn the CBPO method. When it came to analysis and write-up, academic partners took the lead but included community partners in iterative consultation about the data analysis and write-up. To ensure that we appropriately credited everyone for their participation, we included community partners as co-authors on academic papers and, as mentioned above, we created a "consortium coauthorship" model that recognizes a much larger number of contributors than is typically the case on academic papers.

The CBPO model was created at the height of the COVID-19 pandemic, responding to colonias residents' request that we not endanger them by visiting in person. However, in hindsight we find that this method worked ideally for a large multi-sited collaborative project, which spans 64 communities in 4 U.S.-Mexico border states. Given the ways the COVID-19 pandemic seems to have permanently changed the way we work and communicate, we would again use the CBPO method if we were

Table 2 Community-Based Participant Observation (CBPO) steps, main elements and expected outcomes

CBPO's Steps	Brief description	Outcome
0. Identifying and engaging community partners	Identify partners (e.g., community leader, NGO community liaison, religious leader, or others) to represent the community to establish a community-researcher relationship that will allow the understanding of community identities and challenges.	Allow researchers to work with partners to discuss the ways and degree of broader community involvement throughout the research process.
1. Aligning research objectives and motivations	Discuss with community partners about their experiences and share the previous research findings to decide the purpose and direction of the project with well-aligned research objectives. The discussions about compensation should start to identify the best way to reward the partnership.	Recognize the partners' time and their role as co-researchers.
2. Co-development of participant observation and fieldnotes protocol	Develop a protocol addressing community partners' questions and feedback for conducting observations and recording field notes. Since community members are the participant observers, the protocol should be clear and in their language(s).	Ensure that processes are equitable, the language is understandable, and the types of observations are acceptable to the community.
3. Recruitment and training	Recruit and train community observers with deep knowledge of and experience in the community and situated knowledge of the research topic. Training sessions include project goals, their tasks, IRB training, and discussions of ethical challenges that are locally relevant.	Provide step-by-step detailed directions for participant observation tasks to record data.
4. Data collection & management	Use a secure platform to collect and store data. Tracking the record of observations with participant I.D.s, allowing time flexibility, and accommodating local needs reinforce support and understanding from community partners.	Show commitment to our partner's time and efforts through flexibility and accommodations to local dynamics. Support the privacy of participants.

Table 2 (continued)

CBPO's Steps	Brief description	Outcome
5. Data analysis	Involve community observers, representatives, and/or community partners in the analysis process or invite them to reflect on the initial findings depending on their interests and availability. These meetings allow community members to propose alternative analyses, and researchers should be open to redoing the analysis.	Ensure community members' space to speak honestly that can correct or contextualize observations and support better research findings that impact their livelihoods.
6. Reporting back	Findings must be reported via different modes of dissemination that are understandable to the community and created in conversation with community partners.	Mobilize the dissemination of knowledge as an intervention approach and guard against extractive research practices.

to re-implement this study; we would, however, supplement it with site visits and community meetings, as is typical in the design of a PAR study.

3.4 Actionability, Significance & Reflexivity

A feature that Participatory Convergence shares with PAR is the extent to which work generates new ideas that guide action in response to social needs (Bradbury 2010). Teams engage in cycles of action and reflection. In the action phase, team members test ideas and gather information. In the reflection phase, they work together to make sense of the data and plan further actions (Reason and Bradbury 2008).

3.4.1 Reflexivity

Reflexibility is the ability of the team to locate themselves as agents of change (Bradbury 2010) through self-awareness, reflecting on power dynamics, and sense-making during the research process. As a team, we all come from different positions and constantly engage to move our agenda forward.

Our diverse backgrounds posed a few challenges at the start of our collaboration. We had to deal with differences in language and varying perspectives and understandings of the issue we were trying to tackle, as well as conflicts between the funder's deadlines and our project goals. Additionally, we had to adapt our research methods—such as face-to-face participatory and survey activities—to adhere to COVID-19 safety protocols. This required us to redesign some parts of our research. These challenges gave us a unique opportunity to reflect on how our diverse expertise and knowledge influenced our project's motivations. As noted by a community partner:

> Over the two years of this project, I've seen an extreme shift in how all team members understand each other – and most especially in how the researchers at ASU [Arizona State University] have come to understand the importance of listening to community members and what they believe are their most pressing needs and desires for themselves and their communities. The team has created an essential space for learning and every team member has made an effort to shift their worldview, which is no easy feat.

Cyclical opportunities for reflection and action enabled us to re-evaluate our work constantly. Our team had regular meetings to discuss our work and receive feedback from each other. We also valued input from our community partners and ensured that their concerns were considered when making project decisions. For example, after the engineering team analyzed water quality samples, we received feedback from the community about the cost of replacing water filters. To address this concern, the engineers identified affordable materials easily accessible to residents. They also developed a STEM educational module for high-school students to teach them how to measure metals in their water and within POU devices to demonstrate that the filters have removed harmful metals. This method was validated by using state-of-the-art instruments and

techniques to ensure its safety. By participating in this educational activity, residents could benefit from discounted filter replacement costs and learn about the potential health risks associated with hazardous metals in their drinking water.

3.4.2 Actionability & Significance

Participatory Convergence projects aim to create meaningful and relevant actions for the flourishing of individuals, communities, and the environment beyond immediate contexts (Bradbury 2010). Actions may span a spectrum of scales, ranging from minor adjustments to substantial structural transformations (Cornish et al. 2023). As we have described above, one of the most significant features of our project is the active involvement and Convergence of academics, partners, and local residents. This allowed us to establish the project objectives and achieve significant actions.

Our shared objectives were not aligned when Action for Water Equity was initiated. Although we all agreed that water insecurity was a significant issue that needed to be addressed, we had varying opinions regarding the underlying causes and the most appropriate solution. The perspectives of engineering, law, anthropology, economics, community partners, and data science resulted in vastly different interpretations of the causes (e.g., racial discrimination vs. economies of scale) and solutions (e.g., installing water filters, political protest, building pipelines to nearby communities, or using social media to connect people with and without access to water). Our Convergence Research aimed to facilitate iterative discussions and better understand the problem, its causes, and potential solutions until we arrived at a shared understanding and vision. In doing so, we also intended to develop a shared and cross-disciplinary methodology, such as utilizing insights from one social science (anthropology) to implement a methodology from another discipline (e.g., choice modeling from marketing and economics).

In keeping with the usual approach in participatory research, our team acknowledged that an intriguing fundamental research problem may not be the most pressing or critical issue for stakeholders. To tackle this, we engaged in repeated conversations with our community partners to determine the problem, its underlying causes, and possible sustainable solutions adapted to Colonia's residents' needs and socioeconomic conditions. As described by one of our members:

> As the team lead for the engineering team, I realized that point-of-use technologies could be an immediate but temporary (3-8 year) solution to clean water in colonias until public water infrastructure was designed, financed, and installed. However, I realized that our community engagement partner has concerns that any temporary solution decreases the urgency for a longer term, permanent solution - and thus may not actually be the right route.

4 Conclusions

The climate crisis and the associated challenges, such as water insecurity, pose significant threats to humanity, which require transdisciplinary solutions. This paper introduces "Participatory Convergence" as a framework to address complex

societal issues. It combines Convergence Research with Participatory Action Research (PAR) and bridges the gaps in the Convergence literature. Convergence Research and PAR intersect powerfully to foster knowledge production and critical awareness of reality. Participatory Convergence provides a holistic approach that actively involves diverse stakeholders in collaborative problem-solving based on the principles of PAR.

The journey from Convergence to Participatory Convergence involves a transition from traditional disciplinary boundaries towards a focus on systems thinking, solutions orientation, and community engagement. Participatory Convergence emphasizes problem-based research, deep integration across disciplines, active partnerships with community stakeholders, actionable and reflexive practices, and the pursuit of significant and meaningful actions. By engaging in these partnerships and collaborations, scholars aim to gain a more holistic understanding of the complex long-term issues and potential solutions (Iwaniec et al. 2020; Moewaka Barnes et al. 2021; Peek et al. 2020; Sheppard et al. 2011; Sillitoe 2004).

Illustrated through the Action for Water Equity project, this paper showcases the application of Participatory Convergence to address the critical issue of water insecurity in U.S. colonias. The project takes a problem-based approach to tackling water insecurity, disproportionately affecting marginalized populations in these under-resourced communities. The project demonstrates deep integration and partnership building by actively involving scholars from various disciplines, community partners, and residents. Participatory Convergence enables researchers, practitioners, and local residents to foster trust, establish a common language, and gain a broader understanding of challenges and potential solutions (Sundstrom et al. 2023). To illustrate, in our project through Participatory Convergence, our team has been able to gain a comprehensive understanding of the issue of water insecurity in Colonias from diverse perspectives. This approach has enabled us to engage in productive and significant discussions and develop tailored solutions that address the specific needs of residents in these communities.

Furthermore, the framework is actionable and reflexive, adapting to challenges such as the COVID-19 pandemic and diverse perspectives within the team. This adaptability ensures that the research remains relevant and impactful. The significant actions taken within the project, including community involvement, data analysis, and the development of novel solutions, exemplify the core principles of Participatory Convergence.

In a broader context, the Participatory Convergence framework serves as a model for addressing pressing global challenges by promoting interdisciplinary collaboration, community engagement, and a commitment to meaningful change. It underscores the importance of embracing complexity, inclusivity, and adaptability in research and action. The Action for Water Equity project provides a practical illustration of how Participatory Convergence can be applied to address the urgent societal issues of our time (e.g., climate change, food insecurity, energy injustices, water insecurity, etc.). By fostering Convergence between diverse disciplines and communities, we can work towards a more sustainable and equitable future, particularly in the context of environmental and social crises.

Acknowledgment The consortium co-authors are Melissa Beresford, Alexandra Brewis, Alireza Farsad, Zhining Gu, Dylan Hendel, Jelena Jankovic-Rankovic, Jiwon Jang, Wendy Jepson, Olga Morales-Pate, John Sabo, Patrick Thomson, Carmen Velasco Rivera, and Madeleine Zheng.

Funding This study is part of the Action for Water Equity project (supported by the U.S. National Science Foundation Growing Convergence Research program (GCR #2021147). Additional funding was from Household Water Insecurity Experiences Research Coordination Network (HWISE RCN, NSF BCS-179972), NSF EEC-1449500: Nanosystems Engineering Research Center on Nanotechnology-Enabled Water Treatment, NSF-SBE 2017491: The Cultural Anthropology Methods Program (CAMP): Advanced Research Training for Ph.D. Students, and NSF BCS-2143766 CAREER: Moral Economies in Water Markets: Implications for Understanding Human Responses to Water Insecurity, Advanced Research Training for Ph.D. Students, JobsOhio SFPN_2020_486365-NAR and the JPB Foundation, and Arizona Water for All, part of the Arizona Water Innovation Initiative supported by the State of Arizona.

Declarations

Conflict of interest The authors declare there is no financial interests or personal relationships influencing this work.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

Apgar, J. Marina., Alejandro Argumedo, and Will Allen. 2009. Building Transdisciplinarity for Managing Complexity: Lessons from Indigenous Practice. *The International Journal of Interdisciplinary Social Sciences: Annual Review* 4(5): 255–270. https://doi.org/10.18848/1833-1882/CGP/v04i05/52925.

Arnstein, Sherry R. 1969. A Ladder Of Citizen Participation. *Journal of the American Planning Association* 35(4): 216–224. https://doi.org/10.1080/01944366908977225.

Bainbridge, William Sims, and Mihail C. Roco, eds. 2016. *Handbook of Science and Technology Convergence*. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-319-07052-0.

Banks, S., and M. Brydon-Miller. 2018. *Ethics in Participatory Research for Health and Social Well-Being: Cases and Commentaries*. Routledge & CRC Press.

Barnes, Moewaka Helen, Garth Harmsworth, Gail Tipa, Wendy Henwood, and Tim McCreanor. 2021. Indigenous-Led Environmental Research in Aotearoa New Zealand: Beyond a Transdisciplinary Model for Best Practice, Empowerment and Action. *AlterNative: an International Journal of Indigenous Peoples* 17(2): 306–316. https://doi.org/10.1177/117718012110193397.

Bernard, H. Russell. 2017. *Research Methods in Anthropology: Qualitative and Quantitative Approaches*. Rowman & Littlefield.

Boser, Susan. 2007. Power, Ethics, and the IRB: Dissonance Over Human Participant Review of Participatory Research. *Qualitative Inquiry* 13(8): 1060–1074. https://doi.org/10.1177/1077800407308220.

Bradbury, Hilary. 2010. What Is Good Action Research?: Why the Resurgent Interest? *Action Research* 8(1): 93–109. https://doi.org/10.1177/1476750310362435.

Bradbury, Hilary, Steve Waddell, Karen O' Brien, Marina Apgar, Ben Teehankee, and Ioan Fazey. 2019. A Call to Action Research for Transformations: The Times Demand It. *Action Research* 17(1): 3–10. https://doi.org/10.1177/1476750319829633.

Brown, Erik, Igor Cavallini Johansen, Ana Paula Bortoleto, Yadu Pokhrel, Suyog Chaudhari, Anthony Cak, Samer Sulaeman, et al. 2022. Feasibility of Hybrid In-Stream Generator-Photovoltaic Systems for Amazonian off-Grid Communities. *PNAS Nexus* 1(3): pga077. https://doi.org/10.1093/pnasnexus/pgac077.

Brown, Joe, Charisma S. Acey, Carmen Anthonj, Dani J. Barrington, Cara D. Beal, Drew Capone, Oliver Cumming, et al. 2023. The Effects of Racism, Social Exclusion, and Discrimination on Achieving Universal Safe Water and Sanitation in High-Income Countries. *The Lancet Global Health* 11(4): e606–e614. https://doi.org/10.1016/S2214-109X(23)00006-2.

Bukovsky, Melissa, Seth McGinnis, Lee Kessenich, Linda Mearns, Harry Podschwit, and Alison Cullen. 2022. Future Changes in Simultaneous Megafires over the United States as Projected by NA-CORDEX Simulations, May, EGU22-6135. https://doi.org/10.5194/egusphere-egu22-6135.

Chahin, Jaime. 2005. The Forgotten Americans: A Voice for Colonia Residents. *Children, Youth and Environments* 15(1): 319–331.

Chen, Elizabeth, Cristina Leos, Sarah D. Kowitt, and Kathryn E. Moracco. 2020. Enhancing Community-Based Participatory Research Through Human-Centered Design Strategies. *Health Promotion Practice* 21(1): 37–48. https://doi.org/10.1177/1524839919850557.

Chester, Mikhail, B. Shane Underwood, Braden Allenby, Margaret Garcia, Constantine Samaras, Samuel Markolf, Kelly Sanders, Benjamin Preston, and Thaddeus R. Miller. 2021. Infrastructure Resilience to Navigate Increasingly Uncertain and Complex Conditions in the Anthropocene. *Npj Urban Sustainability* 1(1): 1–6. https://doi.org/10.1038/s42949-021-00016-y.

Cornish, Flora, Nancy Breton, Ulises Moreno-Tabarez, Jenna Delgado, Mohi Rua, Ama de-Graft Aikins, and Darrin Hodgetts. 2023. Participatory Action Research. *Nature Reviews Methods Primers* 3(1): 1–14. https://doi.org/10.1038/s43586-023-00214-1.

Cornwall, Andrea. 2008. Unpacking 'Participation' Models, Meanings and Practices. *Community Development Journal* 43(3): 269–283. https://doi.org/10.1093/cdj/bsn010.

Cornwall, Andrea, and Rachel Jewkes. 1995. What Is Participatory Research? *Social Science & Medicine* 41(12): 1667–1676. https://doi.org/10.1016/0277-9536(95)00127-S.

Díaz-Arévalo, Juan Mario. 2022. In Search of the Ontology of Participation in Participatory Action Research: Orlando Fals-Borda's Participatory Turn, 1977–1980. *Action Research* 20(4): 343–362. https://doi.org/10.1177/14767503221103571.

FalsBorda, Orlando. 1979. Investigating Reality in Order to Transform It: The Colombian Experience. *Dialectical Anthropology* 4(1): 33–55.

FalsBorda, Orlando, and Mohammad Anisur Rahman, eds. 1991. *Action and Knowledge. Breaking the Monopoly with Participatory Action-Research*. Santa Fé de Bogotá: CINEP.

FalsBorda, Orlando. 2006. The North-South Convergence: A 30-Year First-Person Assessment of PAR. *Action Research* 4(3): 351–358. https://doi.org/10.1177/1476750306066806.

Gajary, Lisa C., Shalini Misra, Anand Desai, Dean M. Evasius, Joy Frechtling, David A. Pendlebury, Joshua D. Schnell, Gary Silverstein, and John Wells. 2023. Convergence Research as a 'System-of-Systems': A Framework and Research Agenda. *Minerva*. https://doi.org/10.1007/s11024-023-09503-1.

Gaventa, John, and Andrea Cornwall. 2015. Power and Knowledge. In *The SAGE Handbook of Action Research*, ed. H. Bradbury, Third.

Grant, Stanley B., Harry Zhang, Shantanu V. Bhide, Thomas Birkland, Emily Berglund, Andrea Dietrich, Jennifer Druhan, et al. 2021. Advances in Water Research—Reversing Freshwater Salinization: A Holistic Approach. https://www.advancesinwaterresearch.org/awr/library/item/20210709/3945381/.

Greenwood, Davydd J., and Morten Levin. 2007. *Introduction to Action Research: Social Research for Social Change*, 2nd ed. Thousand Oaks, Calif.: Sage Publications.

Gubrium, Aline, and Krista Harper. 2016. *Participatory Visual and Digital Methods*. New York: Routledge. https://doi.org/10.4324/9781315423012.

Harper, K., A. Bates, V. Ogechi, J. Cantor, M. Cowan, & M. Pineda Shokooh. 2023. Clean, green, and just? Community perspectives on the renewable energy transition in a New England city. *Science Talks*, *6*, 100188. https://doi.org/10.1016/j.sctalk.2023.100188

IPCC. 2022. *Climate Change 2022 – Impacts, Adaptation and Vulnerability: Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge: Cambridge University Press. https://doi.org/10.1017/9781009325844.

Israel, Barbara A., Edith A. Parker, Zachary Rowe, Alicia Salvatore, Meredith Minkler, Jesús López, Arlene Butz, et al. 2005. Community-Based Participatory Research: Lessons Learned from the Centers for Children's Environmental Health and Disease Prevention Research. *Environmental Health Perspectives* 113(10): 1463–1471. https://doi.org/10.1289/ehp.7675.

Iwaniec, David M., Elizabeth M. Cook, Melissa J. Davidson, Marta Berbés-Blázquez, E. Matei Georgescu, Scott Krayenhoff, Ariane Middel, David A. Sampson, and Nancy B. Grimm. 2020. The Co-Production of Sustainable Future Scenarios. *Landscape and Urban Planning* 197(May): 103744. https://doi.org/10.1016/j.landurbplan.2020.103744.

Jacobson, S.K, and J.G Robinson. 2009. Training the New Conservationist: Cross-Disciplinary Education in the 1990s. August. https://library.wcs.org/doi/ctl/view/mid/33065/pubid/PUB13029.aspx.

Jepson, Wendy. 2012. Claiming Space, Claiming Water: Contested Legal Geographies of Water in South Texas. *Annals of the Association of American Geographers* 102(3): 614–631. https://doi.org/10.1080/00045608.2011.641897.

Jepson, Wendy. 2014. Measuring 'No-Win' Waterscapes: Experience-Based Scales and Classification Approaches to Assess Household Water Security in Colonias on the US–Mexico Border. *Geoforum* 51(January): 107–120. https://doi.org/10.1016/j.geoforum.2013.10.002.

Jepson, Wendy, and Heather Lee Brown. 2014. 'If No Gasoline, No Water': Privatizing Drinking Water Quality in South Texas Colonias. *Environment and Planning a: Economy and Space* 46(5): 1032–1048. https://doi.org/10.1068/a46170.

Jepson, Wendy, and Emily Vandewalle. 2016. Household Water Insecurity in the Global North: A Study of Rural and Periurban Settlements on the Texas-Mexico Border. *The Professional Geographer* 68(1): 66–81. https://doi.org/10.1080/00330124.2015.1028324.

Jepson, Wendy, Jessica Budds, Laura Eichelberger, Leila Harris, Emma Norman, Kathleen O'Reilly, Amber Pearson, et al. 2017. Advancing Human Capabilities for Water Security: A Relational Approach. *Water Security* 1(July): 46–52. https://doi.org/10.1016/j.wasec.2017.07.001.

Juarez, Jose G., Ester Carbajal, Katherine L. Dickinson, Selene Garcia-Luna, Nga Vuong, John-Paul. Mutebi, Ryan R. Hemme, Ismael Badillo-Vargas, and Gabriel L. Hamer. 2022. The Unreachable Doorbells of South Texas: Community Engagement in Colonias on the US-Mexico Border for Mosquito Control. *BMC Public Health* 22(1): 1176. https://doi.org/10.1186/s12889-022-13426-z.

Lakhina, Shefali Juneja, Elaina J. Sutley, and Jay Wilson. 2021. 'How Do We Actually Do Convergence' for Disaster Resilience? Cases from Australia and the United States. *International Journal of Disaster Risk Science* 12(3): 299–311. https://doi.org/10.1007/s13753-021-00340-y.

Landes, Laura, Nargish Patwoary, Carmen Velasco, and Madeleine Zheng. 2021. Innovative Research to Improve Water Security in Colonias—A RCAP and Arizona State University Collaboration. *Rural Matters. The Magazine of the Rural Community Assistance Partnership*, 2021.

Lang, Daniel J., Arnim Wiek, Matthias Bergmann, Michael Stauffacher, Pim Martens, Peter Moll, Mark Swilling, and Christopher J. Thomas. 2012. Transdisciplinary Research in Sustainability Science: Practice, Principles, and Challenges. *Sustainability Science* 7(1): 25–43. https://doi.org/10.1007/s11625-011-0149-x.

Li, Wenwen. 2020. GeoAI: Where Machine Learning and Big Data Converge in GIScience. *Journal of Spatial Information Science* 2020(20): 71–77. https://doi.org/10.5311/JOSIS.2020.20.658.

Li, Wenwen, Michael F. Goodchild, and Robert Raskin. 2014. Towards Geospatial Semantic Search: Exploiting Latent Semantic Relations in Geospatial Data. *International Journal of Digital Earth* 7(1): 17–37. https://doi.org/10.1080/17538947.2012.674561.

Meehan, Katie, Jason R. Jurjevich, Nicholas M. J. W. Chun, and Justin Sherrill. 2020. Geographies of Insecure Water Access and the Housing-Water Nexus in US Cities. *Proceedings of the National Academy of Sciences* 117(46): 28700–28707. https://doi.org/10.1073/pnas.2007361117.

Moran, Emilio F., Maria Claudia Lopez, Rachel Mourão, Erik Brown, Aaron M. McCright, Judith Walgren, Ana Paula Bortoleto, et al. 2022. Advancing Convergence Research: Renewable Energy Solutions for off-Grid Communities. *Proceedings of the National Academy of Sciences* 119(49): e2207754119. https://doi.org/10.1073/pnas.2207754119.

National Research Council. 2014. *Convergence: Facilitating Transdisciplinary Integration of Life Sciences, Physical Sciences, Engineering, and Beyond*. Washington, D.C.: National Academies Press. https://doi.org/10.17226/18722.

NSF. 2019. Learn About Convergence Research—Research Approaches. *NSF—National Science Foundation*. National Science Foundation. 2019. https://new.nsf.gov/funding/learn/research-types/learn-about-convergence-research.

O'Rourke, Michael, Stephen Crowley, and Chad Gonnerman. 2016. On the Nature of Cross-Disciplinary Integration: A Philosophical Framework. *Studies in History and Philosophy of Science Part C: Studies in History and Philosophy of Biological and Biomedical Sciences* 56(April): 62–70. https://doi.org/10.1016/j.shpsc.2015.10.003.

Peek, Lori, Jennifer Tobin, Rachel M. Adams, Wu Haorui, and Mason Clay Mathews. 2020. A Framework for Convergence Research in the Hazards and Disaster Field: The Natural Hazards Engineering Research Infrastructure CONVERGE Facility. *Frontiers in Built Environment*. https://doi.org/10.3389/fbuil.2020.00110.

Pohl, Christian, Lorrae van Kerkhoff, Gertrude Hirsch Hadorn, and Gabriele Bammer. 2008. Integration. In *Handbook of Transdisciplinary Research*, eds. Gertrude Hirsch Hadorn, Holger Hoffmann-Riem, Susette Biber-Klemm, Walter Grossenbacher-Mansuy, Dominique Joye, Christian Pohl, Urs Wiesmann, and Elisabeth Zemp. Dordrecht: Springer, Netherlands. https://doi.org/10.1007/978-1-4020-6699-3_27.

Pretty, Jules N. 1995. Participatory Learning for Sustainable Agriculture. *World Development* 23(8): 1247–1263. https://doi.org/10.1016/0305-750X(95)00046-F.

RCAP, CU, RCAC, and CAST. 2015. *U.S.-Mexico Border Needs Assessment and Support Project: Phase II Assessment Report*. Rural Community Assistance Partnership (RCAP); Communities Unlimited (CU); Rural Community Assistance Partnership (RCAC); The Center for Advanced Spatial Technologies (CAST).

Reason, Peter, and Hilary Bradbury. 2008. Introduction. In *The SAGE Handbook of Action Research: Participative Inquiry and Practice, Second*, ed. Fikret Berkes, Johan Colding, and Carl Folke, 1–10. Sage Publications. https://doi.org/10.4135/9781848607934.

Roco, Mihail C., and William Sims Bainbridge, eds. 2003. *Converging Technologies for Improving Human Performance*. Dordrecht: Springer, Netherlands. https://doi.org/10.1007/978-94-017-0359-8.

Roco, Mihail C. 2016. Principles and Methods That Facilitate Convergence. In *Handbook of Science and Technology Convergence*, ed. William Sims Bainbridge and Mihail C. Roco, 17–41. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-319-07052-0_2.

Roco, Mihail C., and William Sims Bainbridge. 2013. The New World of Discovery, Invention, and Innovation: Convergence of Knowledge, Technology, and Society. *Journal of Nanoparticle Research* 15(9): 1946. https://doi.org/10.1007/s11051-013-1946-1.

Roco, Mihail C., William Sims Bainbridge, Bruce Edward Tonn, and George Whitesides, eds. 2013. *Convergence of Knowledge, Technology and Society: Beyond Convergence of Nano-Bio-Info-Cognitive Technologies. Science Policy Reports*. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-319-02204-8.

Roller, Zoë., N. Stephen Gasteyer, W. Lai. Nelson, and M. Shingne. 2019. *Closing the Water Access Gap in the United States: A National Action Plan*. Oakland, CA: Dig Deep and US Water Alliance.

Roque, Anais, Amber Wutich, Barbara Quimby, Sarah Porter, Madeleine Zheng, Mohammed Jobayer Hossain, and Alexandra Brewis. 2022. "Participatory Approaches in Water Research: A Review. *Wires Water*. https://doi.org/10.1002/wat2.1577.

Roque, Anais, Amber Wutich, Alexandra Brewis, Melissa Beresford, Laura Landes, Olga Morales-Pate, Ramon Lucero, et al. 2023. Community-Based Participant-Observation (CBPO): A Participatory Method for Ethnographic Research. *Field Methods*. https://doi.org/10.1177/1525822X231198989.

Sheppard, Stephen R J., Alison Shaw, David Flanders, Sarah Burch, Arnim Wiek, Jeff Carmichael, John Robinson, and Stewart Cohen. 2011. Future Visioning of Local Climate Change: A Framework for Community Engagement and Planning with Scenarios and Visualisation. *Futures* 43(4): 400–412. https://doi.org/10.1016/j.futures.2011.01.009.

Sillitoe, Paul. 2004. Interdisciplinary Experiences: Working with Indigenous Knowledge in Development. *Interdisciplinary Science Reviews* 29(1): 6–23. https://doi.org/10.1179/030801804225012428.

Stock, Paul, and Rob J. F. Burton. 2011. Defining Terms for Integrated (Multi-Inter-Trans-Disciplinary) Sustainability Research. *Sustainability* 3(8): 1090–1113. https://doi.org/10.3390/su3081090.

Stoler, Justin, Wendy Jepson, Amber Wutich, Carmen A. Velasco, Patrick Thomson, Chad Staddon, and Paul Westerhoff. 2022. Modular, Adaptive, and Decentralised Water Infrastructure: Promises and Perils for Water Justice. *Current Opinion in Environmental Sustainability* 57(August): 101202. https://doi.org/10.1016/j.cosust.2022.101202.

Sundstrom, Shana M., David G. Angeler, Jessica G. Ernakovich, Jorge H. García, Joseph A. Hamm, Orville Huntington, and Craig R. Allen. 2023. The Emergence of Convergence. *Elementa: Science of the Anthropocene* 11(1): 00128. https://doi.org/10.1525/elementa.2022.00128.

Thomas, Nigel, and Claire O'Kane. 1998. The Ethics of Participatory Research with Children. *Children & Society* 12(5): 336–348. https://doi.org/10.1111/j.1099-0860.1998.tb00090.x.

Vandewalle, Emily, and Wendy Jepson. 2015. Mediating Water Governance: Point-of-Use Water Filtration Devices for Low-Income Communities along the US-Mexico Border: Mediating Water Governance. *Geo: Geography and Environment* 2 (2): 107–121. https://doi.org/10.1002/geo2.9.

Vaughn, Lisa M., and Farrah Jacquez. 2020. Participatory Research Methods—Choice Points in the Research Process. *Journal of Participatory Research Methods*. https://doi.org/10.35844/001c.13244.

Westerhoff, Paul, Amber Wutich, and Curt Carlson. 2021. Value Propositions Provide a Roadmap for Convergent Research on Environmental Topics. *Environmental Science & Technology* 55(20): 13579–13582. https://doi.org/10.1021/acs.est.1c05013.

Wickson, F., A.L. Carew, and A.W. Russell. 2006. Transdisciplinary Research: Characteristics, Quandaries and Quality. *Futures* 38(9): 1046–1059. https://doi.org/10.1016/j.futures.2006.02.011.

Wilson, N. 2019. On the Road to Convergence Research. *BioScience* Oxford Academic 2019. https://academic-oup-com.ezproxy1.lib.asu.edu/bioscience/article/69/8/587/5530927.

Wutich, Amber, and Kathleen Ragsdale. 2008. Water Insecurity and Emotional Distress: Coping with Supply, Access, and Seasonal Variability of Water in a Bolivian Squatter Settlement. *Social Science & Medicine* 67(12): 2116–2125. https://doi.org/10.1016/j.socscimed.2008.09.042.

Wutich, Amber, Wendy Jepson, Carmen Velasco, Anais Roque, Gu Zhining, Michael Hanemann, Mohammed Jobayer Hossain, et al. 2022. Water Insecurity in the Global North: A Review of Experiences in U.S. Colonias Communities along the Mexico Border. *Wires Water*. https://doi.org/10.1002/wat2.1595.

Wutich, Amber, Patrick Thomson, Wendy Jepson, Justin Stoler, Alicia D. Cooperman, James Doss-Gollin, Anish Jantrania, et al. 2023. MAD Water: Integrating Modular, Adaptive, and Decentralized Approaches for Water Security in the Climate Change Era. *Wires Water* 10(6): e1680. https://doi.org/10.1002/wat2.1680.

Zheng, Madeleine, Amber Wutich, Alexandra Brewis, and Stavros Kavouras. 2022. Health Impacts of Water and Sanitation Insecurity in the Global North: A Scoping Literature Review for U.S. Colonias on the Mexico Border. *Journal of Water and Health* 20(9): 1329–1342. https://doi.org/10.2166/wh.2022.085.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.